Reproducibility and Uncertainty in Radiated Emission Measurements at Open Area Test Sites and in Semianechoic Chambers Using a Spherical Dipole Radiator

Tae-Weon Kang and Hyo-Tae Kim

***Abstract*—Radiated emission (RE) measurements have been made in 3 m semianechoic chambers (SACs) and at 10 m open area test sites (OATSs) using a spherical dipole radiator (SDR) as a reference emitter. To compare the electric field strengths of the SACs with those of the OATSs, the analytical extrapolation factor for the SDR is employed. For the experiments, 23 discrete frequencies are chosen in the frequency range of 30 to 1000 MHz. The resultant reproducibility (2σ) is obtained to be 7.3 and 6.0 dB for the SAC and the OATS measurements. Accounted possible contributions to overall uncertainty, the expanded uncertainty ($k = 2$) is evaluated to be 5.1 and 5.3 dB for the SAC and OATS measurements, respectively.**

***Index Terms*—Open area test site, radiated emission, reproducibility, semianechoic chamber, uncertainty.**

I. Introduction

The issue of reproducibility in RE measurements has been addressed by many in the electromagnetic compatibility (EMC) community. Heirman [1] presented a technique for OATS evaluations using a spherical dipole emission source in the frequency range of 30– 300 MHz. A method for finding site anomalies which were indicated by measured data deferring by above 4 dB from the criteria [2], [3] was discussed for four test sites [4]. Halaberda and Rivers [5] compared RE test results collected from 44 sites including both OATSs and SACs for 3- and 10-m distances. In this paper, it was demonstrated that OATS has larger deviations from the mean than the SAC and that over 60% of the sites has one or more frequencies with differences of over 5 dB from the mean. A commercially available emission source with the repeatability of 0.5 dB was used for the measurements in 30–500-MHz frequency range with the increment of 10 MHz [5]. Statistical comparisons of site-to-site measurement reproducibility at 10-m sites were performed in the frequency range of 30–1000 MHz [6] and of 30–2000 MHz [7]. A specially designed emitter capable of generating 101 frequencies spaced logarithmically and an improved signal source of 10 kHz–4 GHz frequency range were used for the two works.

For evaluating the RE measurement reproducibility, it is essential to use a reference emitter with high stability and good repeatability. A commercially available SDR is employed in this paper. In general, if RE measurements are made at test sites of different distances, it is not possible to compare the measurement data in a direct manner. In this case, an appropriate conversion factor accounting the difference of the measurement distance, namely, extrapolation factor can be obtained, provided that the electromagnetic characteristics of the reference emitter are explicitly known. The factor is then applied to measurement data at one distance in order to convert them into those at the other distance.

This paper begins with obtaining an analytical extrapolation factor using the analytic solution of the SDR, i.e., a reference emitter. RE measurements are performed in four 3-m SACs and twelve 10-m OATSs under the condition that the gap voltage of the SDR at each test site is maintained with good repeatability. In order to compare the 3-m SAC data with those of a 10 m OATS, the extrapolation factor obtained analytically is applied to convert the 3 m SAC data into equivalent data for 10 m distance. The converted results are compared with 10 m OATS measurement data and associated measurement uncertainty is also discussed. Finally, the evaluated reproducibility of RE measurements in the SAC and at the OATS is presented.

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II. The Extrapolation Factor of the SDR

In the evaluation of the extrapolation factor, it is assumed that the inner space of the SAC is a simulated ideal OATS even though electromagnetic absorbers lined along the four side walls and the ceiling of the SAC do not completely absorb incident waves. Suitability of an SAC for RE compliance tests should be evaluated by measuring the normalized site attenuation (NSA) over the defined test volume [2], [3]. Among the four SACs in this paper, three SACs meet the NSA requirement, whereas the other one is originally constructed not for RE compliance tests but for scanning the emission frequency because of space limitations.

In this paper, a commercially available SDR with high stability and good repeatability is used to generate a reference signal. It is preferable to compare the measured data with an absolute standard: the absolute standard is not available because an ideal OATS does not exit. Since the OATS is specified as the reference facility for RE measurements [2], [3], the mean of field strengths measured at the OATSs is to be the best alternative [6].

As mentioned previously, the field strength measured in the 3 m SAC is converted to equivalent data at 10 m distance to compare them with the 10 m OATS data. For an actual equipment under test (EUT), extrapolation is made by procedures specified in the product standards [3]. The extrapolation, generally, does not follow the simple inverse distance law. Instead, it highly depends on the radiation pattern of the EUT, the frequency and polarization [3], [8], [9]. Since the spherical radiator whose electromagnetic characteristics are known is employed, a theoretical extrapolation factor can be readily obtained. The electric field of the SDR placed in free space, illustrated in Fig. 1, is given by [10], [11]

$$E_r(r,\theta) = \left(\frac{2k_0}{\pi}\right)^{1/2} \frac{1}{j\omega\epsilon_0 r} \sum_{n=1}^{\infty} (n+1) A_n h_n^{(2)}(k_0 r) \cdot \left\{\cot\theta P_n^1(\cos\theta) - \csc\theta P_{n-1}^1(\cos\theta)\right\} \quad (1)$$

$$E_\theta(r,\theta) = \left(\frac{2k_0}{\pi}\right)^{1/2} \frac{1}{j\omega\epsilon_0 r} \sum_{n=1}^{\infty} A_n P_n^1(\cos\theta) \cdot \left\{n h_n^{(2)}(k_0 r) - k_0 r h_{n-1}^{(2)}(k_0 r)\right\} \quad (2)$$

$$A_n = \left(\frac{\pi}{2k_0}\right)^{1/2} \frac{2n+1}{2n(n+1)} \cdot \frac{j\omega\epsilon_0 V P_n^1(0)}{n h_n^{(2)}(k_0 a) - k_0 a h_{n-1}^{(2)}(k_0 a)} \quad (3)$$

where $P_n^1(\cos\theta)$ an associated Legendre function of the first kind of order n and degree 1

$$h_n^{(2)}(k_0 r) = \left(\frac{\pi}{2k_0 r}\right)^{1/2} H_{n+\frac{1}{2}}(k_0 r) = j_n(k_0 r) - j\, y_n(k_0 r) \quad (4)$$

the second kind spherical Hankel function of order n:

- k_0 free space wavenumber;
- ϵ_0 permittivity of free space;
- ω angular frequency;
- a the radius of the SDR;

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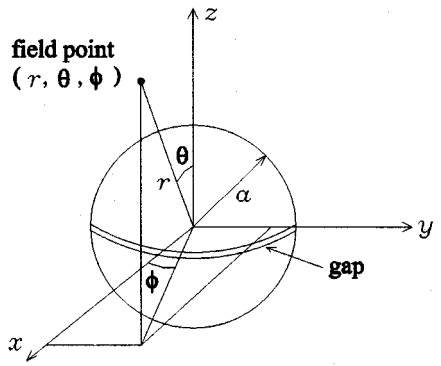


Fig. 1. The spherical dipole radiator and its coordinates.

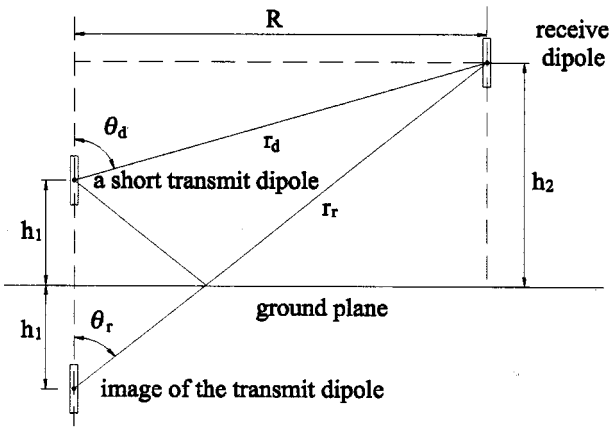


Fig. 2. The equivalent situation of the short dipole at an OATS for vertical polarization.

r the distance from the center of the SDR to the field point;
V the gap voltage between the two hemisphere radiating elements.

Since the SDR has the diameter of 10 cm which is relatively small compared with the distance of 3 m or 10 m, it can be considered as an electrically short dipole or an infinitesimal dipole in the frequency range of interest. In order to investigate closeness of characteristics of the SDR to those of the short dipole, the behavior of the short dipole is briefly considered. The electric field components of a short dipole oriented along the z-axis, carrying the current I_0 in free space are given by [12]

$$E_r = \eta_0 \frac{I_0 L \cos\theta}{2\pi r^2}\left(1+\frac{1}{jk_0 r}\right)e^{-jk_0 r} \tag{5}$$

$$E_\theta = j\eta_0 \frac{k_0 I_0 L \sin\theta}{4\pi r}\left(1+\frac{1}{jk_0 r}+\frac{1}{(jk_0 r)^2}\right)e^{-jk_0 r} \tag{6}$$

where L and η_0 are the length of the dipole and the intrinsic impedance of free space, respectively. Here, the terms of $1/jk_0r$ and $1/(jk_0r)^2$ in the parenthesis of (6) and E_r in (5) are referred to as near-field (NF) terms because they do not produce radiated power. Instead, they cause induced currents in wires placed in close proximity to the transmit dipole. The term of 1 in the parenthesis of (6) is referred to as a far-field (FF) term since the term contributes to the radiated power.

In order to calculate the field strength at an OATS, it is assumed that the ground plane of the OATS is perfectly conducting. Keeping the gap voltage of the SDR and the current flowing along the short dipole a constant, the electric field strength at the OATS is calculated at the distance of 3 m and 10 m. Fig. 2 is the equivalent situation of the short dipole at the OATS for vertical polarization. When RE measurements are performed at an OATS the receive antenna is positioned both in horizontal and vertical polarizations. Therefore, one only needs to compute the parallel component of the radiated electric field at the receive antenna. It is assumed that the receive antenna is short compared with λ so that the antenna does not disturb the field to be measured and the parallel contribution of the electric field at points along its length is the same as its value at a point corresponding to the center of the antenna. Moreover, there is no mutual coupling between the SDR or the short dipole and the receive dipole. Referring to Fig. 2, the parallel component of the electric field at the receiving position for a z-directed short dipole is given by [13]

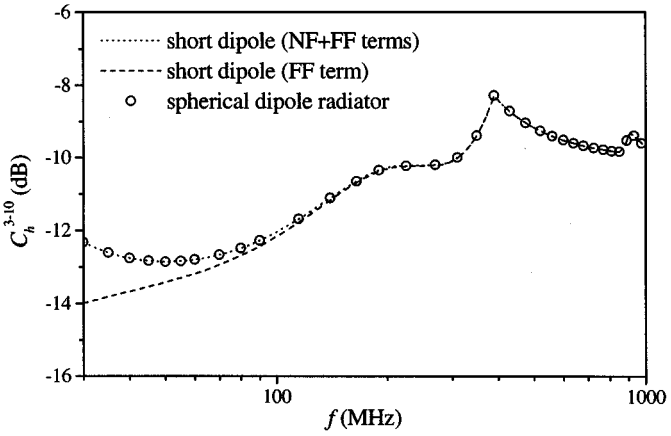
$$E_{||}(r,\theta) = \cos\theta E_r(r,\theta) - \sin\theta E_\theta(r,\theta). \tag{7}$$

Note that the angle of θ depends on the height of transmit and receive antennas for vertical polarization whereas it is a constant of $\pi/2$ for horizontal. Using the simple image theory for accounting the ground plane of the OATS, the total field composed of direct and ground plane-reflected waves can be easily obtained.

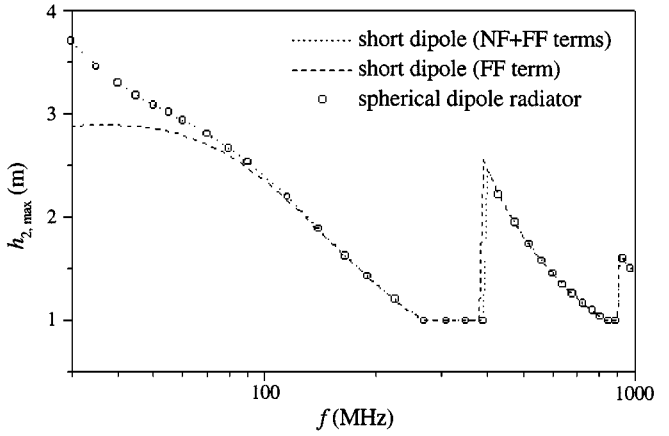
In order to maximize the electric field strength at an OATS, the receive antenna is scanned in the height range of 1–4 m for the distance of 3 m and 10 m. Subtracting the 3 m field strength from the 10 m value at the OATS for each frequency gives the extrapolation factor

$$C^{3-10} = E^{10\,\mathrm{m}}_{\mathrm{OATS}} - E^{3\,\mathrm{m}}_{\mathrm{OATS}} \tag{8}$$

where $E^{3\,\mathrm{m}}_{\mathrm{OATS}}$ and $E^{10\,\mathrm{m}}_{\mathrm{OATS}}$ are the electric field strengths in dBμV/m calculated at the OATS for the corresponding distance. In the calcula-



(a)



(b)

Fig. 3. (a) The 3- to 10-m extrapolation factors and (b) the receiving height at which the received field strength is maximized at a 3 m OATS. They are for horizontal polarization: $h_1 = 0.875$ m, $h_2 = 1$–4 m.

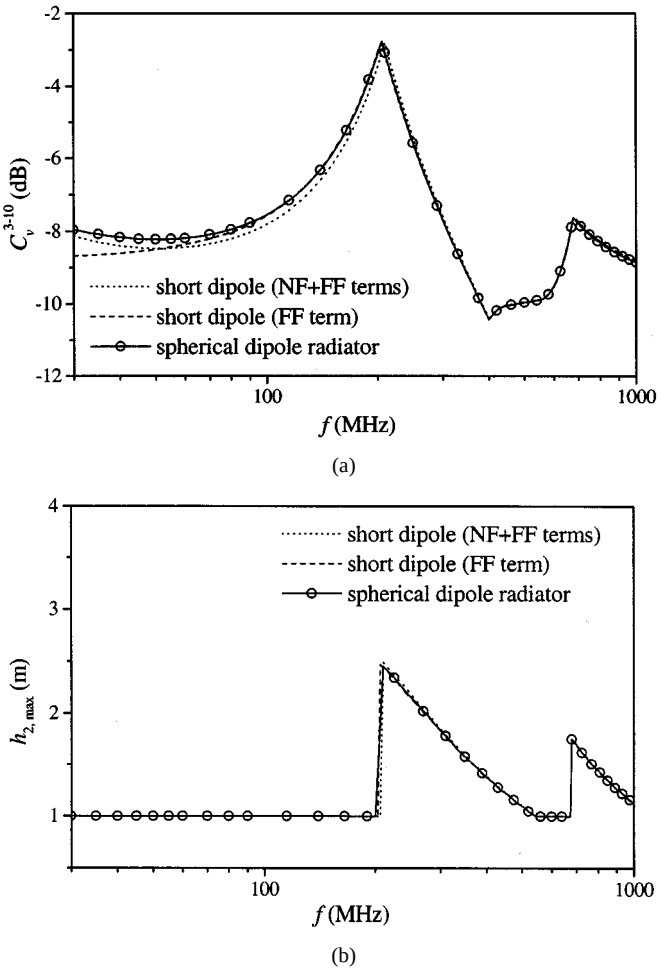


Fig. 4. (a) The 3 m to-10 m extrapolation factors and (b) the receiving height at which the received field strength is maximized at a 3 m OATS. They are for vertical polarization: $h_1 = 0.875$ m, $h_2 = 1$–4 m.

tion of the extrapolation factor, the receive antenna is scanned with the increment of 0.5 cm. Note that the field strengths in (8) are calculated analytically using (1)–(7).

Fig. 3 depicts the calculated extrapolation factors of the SDR and the corresponding receiving height $h_{2,\max}$ at which the received field strength for 3 m distance is maximized for horizontal polarization. As shown in Fig. 3, the factors of the SDR and the short dipole are identical, provided that both the FF- and NF-terms of the field expressions in (5) and (6) for the short dipole are accounted. This is because the receiving point is in the boresight of the transmitting source and its image regardless of the receiving height, i.e., $\theta = \pi/2$ in (7). If the NF-term of the short dipole is not included as in the calculation of the NSA [2], [13], the difference in the extrapolation factor between the SDR and the short dipole amounts to 1.8 dB. The higher the frequency becomes, the larger the electrical distance between the transmit and receive antennas is. This results in good agreement of the two factors. It is of interest to note that the extrapolation factors and the corresponding heights $h_{2,\max}$'s of the short dipole and the SDR are exactly the same. Therefore, the SDR can be considered as a short dipole for horizontal polarization. Moreover, the maximum of $h_{2,\max}$ for the SDR is below 3.7 m in the entire range of frequency.

Fig. 4 is the extrapolation factor and corresponding receive antenna height for vertical polarization. Not only the extrapolation factor but the height of the two transmitting sources show little difference. It is found that $h_{2,\max}$ of the SDR reaches up to 2.5 m and that the behavior of the height versus frequency is nearly invariant with the source type. From Figs. 3(b) and 4(b), $h_{2,\max}$ is less than the maximum height of the scan range, 4 m. This is important because one SAC among the four SACs has the feasible scan range of 1–3.5 m.

A simple addition of the extrapolation factor to the field strengths measured in the 3 m SAC gives the equivalent 10 m SAC data. Consequently, we are now capable of comparing the absolute field strengths of the SAC and the OATS at the same distance of 10 m. It should be noted that the extrapolation factor described in this section is only applicable for an SDR and therefore not usable to a general EUT.

III. Measurement

In this section, the reference emitter employed in this study and the measurement setup are described.

A. The Reference Emitter

A commercially available SDR, SRS2100, Seth Corporation, USA, with the repeatability of 0.2–0.5 dB has been used as a reference emitter. Its frequency range is 10 MHz to 2.5 GHz. The SDR system consists of a spherical radiator with a 10 cm outer diameter and a control unit to interface to the operator and an external RF signal source. The radiator is electrically isolated from the RF source by optical fiber cables. The use of the optical fiber cables between the control unit and the radiator eliminates the disruption of radiated field to be measured. Details on the SDR can be found in [11].

Before the comparison test commences, the RF input power for each frequency is determined in advance at the pilot site so that subsequent gap voltage of the SDR is in typical useful range of 0.2–1.0 V. Moreover, during the procedure of determining the gap voltage care should be taken not to saturate the RF amplifier installed in the spherical radiator due to high RF input power. The gap voltage is determined for both horizontal and vertical polarizations and is referred to as a reference gap voltage as illustrated in Fig. 5. The reference gap voltages for the two polarizations were nearly identical except at the frequency of 250 MHz.

In order to maintain the radiated power as close as possible at each test site, the gap voltage of the SDR has been monitored within $+12/-8$ mV and $+36/-22$ mV with respect to the reference gap voltage for horizontal and vertical polarizations, respectively. Therefore, the largest variation of the gap voltages affects reproducibility of the reference field generation by a factor of 0.4 dB.

B. Measurement Setup

Fig. 6 shows a typical measurement setup. The SDR is positioned at the center of a turntable of an 0.8-m high rotatable wooden table [2], [3]. Because of the round shape of the radiator, a dielectric support made of high-density Styrofoam is used to fix the spherical radiator on the flat table. To confirm that the support does not affect the radiated field to be measured, scattering parameters of the support were measured using a network analyzer HP8753C and a coaxial airline in which a machined sample of the dielectric support was installed: $|S_{11}| \leq -47.5$ dB, $|S_{21}| \leq -0.004$ dB in the frequency range of 30–1000 MHz. Since the SDR is an approximated short dipole, it is determined that no rotation of the turntable is necessary.

The wooden table also affects the electric fields to be measured, which has been investigated extensively by an experimental approach [5], [14], [15]. Inclusion of the effect of the table on uncertainty makes associated analysis complicated. Therefore, it is assumed that the effect of the dielectric table is implicitly incorporated into the measured electric field and subsequently the influence is accounted by measurement reproducibility which will be discussed later. The total height of the radiator above the metal ground plane is 0.875 m.

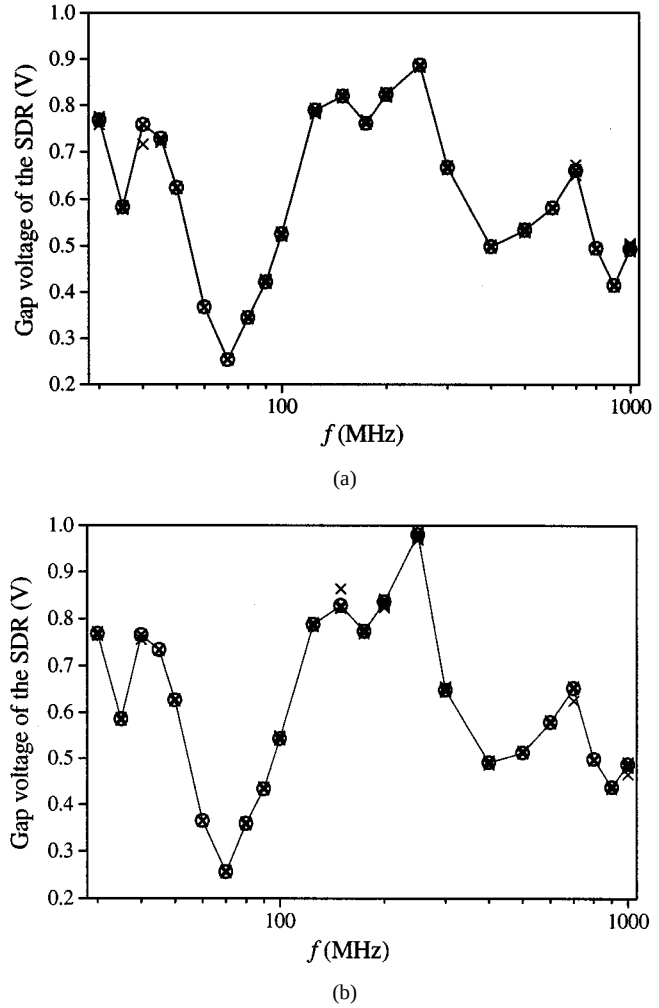


Fig. 5. The gap voltage of the SDR (a) for horizontal and (b) for vertical polarizations. -○- for the reference gap voltage. × for the gap voltage monitored at 4 SACs and 12 OATSs.

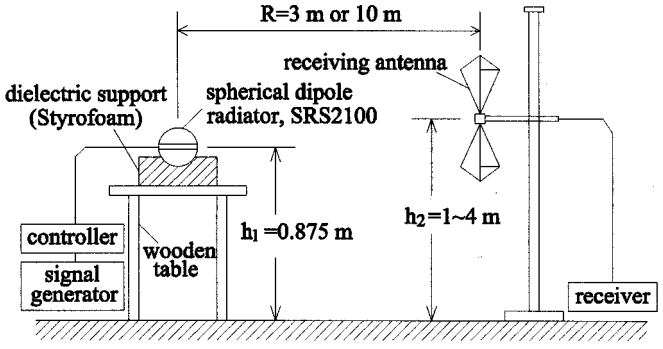


Fig. 6. Measurement setup.

Measurement frequencies are the 23 discrete frequencies selected from the frequency list of ANSI C63.4 [2]. Measurements are made at 3 m distance in the SACs and at 10 m distance at the OATSs for both horizontal and vertical polarizations. For vertical measurements, the SDR is aligned on the wooden table so that its equator made of Teflon is parallel to the ground plane, whereas for horizontal measurements the radiator is rotated about 90 degrees where the rotation axis coincides with the measurement axis defined as a horizontally interconnecting line between the center of the SDR and the receive antenna. The receive antenna is scanned 1 to 4 m and maximum field strengths are recorded.

TABLE I
NUMBER OF TEST SITES AT THE MEASUREMENT FREQUENCIES

Open area test site	Semi-anechoic chamber
Horizontal polarization:	
$n = 5$; 100 MHz	
$n = 10$; 90 MHz	
$n = 11$; 30 MHz	For both polarizations
$n = 12$; elsewhere	$n = 3$
Vertical polarization:	
$n = 5$; 100 MHz	
$n = 10$; 90 MHz	
$n = 12$; elsewhere	

The receive antenna can be a set of biconical and log-periodic antennas, or a bilog antenna to cover the full frequency range of interest. An electromagnetic interference (EMI) receiver capable of measuring the electric field strengths is used. The receiver is used in the quasipeak mode with 120 kHz-bandwidth.

The antenna factor (AF) of the receive antenna should be added to reading of the EMI receiver to convert it into electric field strength. Although the AF is highly dependent on the polarization and on height above a ground plane as discussed in [16]–[19], the free space AF is used and the variation of AF with height is considered as a contribution to measurement uncertainty. The loss of RF cables used for instrumentation also should be compensated.

For the RE measurements, the SDR system has been carried from site to site to make on-site measurement. At each site, the measurement system is configured in accordance with international EMI regulations [2], [3]. Because of limited battery duration of the SDR, the measurement for the selected frequencies has been made approximately in two hours at each site. The measurement can be regarded as a blind proficiency test in view of only the amplitude in RE tests, as the measurement frequency is explicitly known to each operator of the test sites.

IV. REPRODUCIBILITY AND UNCERTAINTY IN RE MEASUREMENTS

A. Reproducibility of Measurements

In order to define reproducibility, the electric field strengths measured at each site are considered as those obtained from repeated measurements. We define the standard uncertainty, or reproducibility for repeated measurements as follows. With n-independent single measurements, the estimated value of x_i of the true value of X_i is given by the arithmetic mean of the individual measured values $v_{i,j}$ $(j = 1, \ldots, n)$

$$x_i = \bar{v_i} = \frac{1}{n}\sum_{j=1}^{n} v_{i,j} \tag{9}$$

where $v_{i,j}$ denotes the field strength of i-th frequency at j-th test site. The estimated standard deviation s_{v_i} of the distribution of the measured values v_i is

$$s_{v_i} = u(v_i) = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(v_{i,j} - \bar{v_i})^2}. \tag{10}$$

Table I shows the number of sites at the measurement frequencies. Because of ambient noise, n is down to 5 for the OATS. To obtain the standard deviation of the mean $\bar{v_i}$, denoted by s_{x_i}, s_{v_i} is divided by $\sqrt{n}$. However, since the number of sites between the OATS and SAC is apparently different as observed in Table I, we will not proceed to

obtain s_{x_i}. Instead, twice the s_{v_i} at each frequency is referred to as reproducibility of RE measurements at the i-th measurement frequency and denoted by 2σ.

It should be emphasized that the gap voltage variation shown in Fig. 5 contributes not to the RE measurement uncertainty of a specific site but to the measurement reproducibility in the intersite comparison. The repeatability of the SDR given by the manufacturer must be considered to evaluate the uncertainty in RE measurements of a site itself. On the other hand, the gap voltage variation influences the reproducibility of the intersite comparison. Moreover, note that the gap voltage variation has already been incorporated into the measurement reproducibility described above. The reason is that the gap voltage variation affects linearly the electric field to be measured because the reference gap voltage has been determined in the linear operating range of the SDR. In the case of extremely large variation of the site-to-site gap voltage, the intersite comparison becomes meaningless.

B. Uncertainty in RE Measurements

Many articles [20]–[26] dealing with the uncertainty in EMC measurements have appeared in recent years. In this paper, we evaluate the uncertainty in RE measurements in accordance with the ISO guide to the expression of uncertainty in measurement [27].

Since the instrumentation is different from site to site, so is the associated measurement uncertainty of each site, though not noticeably. We deal with each contribution to uncertainty by the same order shown in Table II in which uncertainty in RE measurements at the distance of 10 m for a specific OATS among the 12 sites is summarized. Since the numbers of the last two columns in Table II are not divided by the weighting factor for the specified probability distribution function, the corresponding numbers are referred to as tolerance. Note that the weighting factors are $\sqrt{3}$ for rectangular, $\sqrt{2}$ for U-shaped, $\sqrt{6}$ for triangular, 2 for normal, and 1 for standard deviation.

To evaluate uncertainty in RE measurements, contributions to uncertainty are grouped into three parts; transmitting and receiving parts, and the test site itself. For the sake of completeness, we briefly describe each contribution and details on each contribution are found in [20]–[26]. Explicit numerics appearing in the following are for 10 m distance in case of no special comments.

- Transmitting part: The absolute level accuracy of the RF signal generator and the repeatability of the SDR are obtained from the manufacturer's specification. Next contribution to uncertainty is mismatch between the control unit of the SDR and the RF signal generator. The mismatch uncertainty in the power transfer between a generator and a load is

$$u_{\mathrm{mis}} = \frac{20\log\left(1 \pm |\Gamma_{\mathrm{G}}||\Gamma_{\mathrm{L}}|\right)}{\sqrt{2}} \tag{11}$$

  where Γ_{G} and Γ_{L} are the reflection coefficients of the signal generator and the load, respectively, and U-shaped distribution is assumed. To obtain the mismatch uncertainty shown in Table II, the value of voltage standing wave ratio (VSWR), 2:1, is used for the signal generator and the control unit of the SDR.
- Receiving part: The AF in 30–1000 MHz-frequency range is calibrated by the Reference Antenna Method (RAM) at 10 m separation distance where the reference antenna is a set of dipole antennas. The uncertainty of the AF calibration is 1.5 dB $(k = 2)$ and details on the RAM are found in [28]. The uncertainty of cable loss measurement is 0.3 dB where the coverage factor is 2.

  The AF of an antenna varies with the antenna height because of mutual coupling with the ground plane. This effect on RE measurements is not specified by manufacturers. The numeric values given in Table II are attainable by numerical simulations [18], [19]. Because of smooth variation of the AF with the frequency for commercially available antennas, it is obvious that the uncertainty due to AF interpolation with frequency is small as shown in Table II.

  The angle of incidence with respect to the antenna boresight decreases as the measurement distance increases. Since the antenna calibration is carried out for the direction of boresight, it is reasonable to assume that the actual field strength could be stronger than the receiver reading by an unknown amount [25]. This influence for vertical polarization is more potent, comparing with that for horizontal polarization, and the contribution to uncertainty is set to be $+0/+0.2$ (horizontal/vertical) and $+0.2/+0.5$ dB for biconical and log-periodic antennas, respectively [21]. Note that the antenna directivity results in an asymmetric uncertainty.

  The phase center of a log-periodic antenna varies with the frequency, which causes uncertainty in RE measurements. Since different kind of the antennas has been used at each site, the uncertainty due to the phase center variation is 0.2–0.3 dB and 0.6–0.9 dB at 10 m and 3 m distances, respectively. The cross-polarization response of a biconical antenna is considered to be negligible, whereas the response of a log-periodic antenna contributes to uncertainty by an amount of 0.9 dB with a rectangular distribution [26]. Since the effect of unbalanced antenna is apparent at low frequency range when the input coaxial cable is aligned parallel to the antenna elements, its contribution to uncertainty can be set as shown in Table II.

  The uncertainty due to mismatch between the receiver and the receive antenna is obtained by (11). In general, the receive antenna is composed of biconical and log-periodic antennas. The associated transition frequency between the two antennas is either 200 or 300 MHz and the field strength measured by the log-periodic antenna is chosen at the transition frequency. A bilog antenna of the frequency range of 30–1000 MHz is also used at several test sites. The VSWR of the bilog antenna deteriorates as the frequency becomes low. If the maximum VSWR is chosen in the full operating frequency range of the bilog antenna, the uncertainty due to mismatch in its relatively high frequency range will be unnecessarily overestimated. For such cases, the transition frequency is arbitrarily set to be 300 MHz to evaluate the uncertainty. The typical VSWR of the receiver under the condition of 0 dB RF attenuation is 2:1 while those of biconical and log-periodic antennas are 4.6:1 and 2.0:1, respectively.

  The uncertainty of the receiver is composed of voltage accuracy, absolute pulse response, pulse repetition rate response, and noise floor proximity. The accuracy of measurement of sine-wave voltages is better than 1–1.5 dB when supplied with a sine-wave signal at a 50 Ω resistive source impedance. Since the transmitting source, Seth SRS2100, produces a continuous-wave (CW) signal, both absolute pulse response and pulse repetition response does not contribute to the combined uncertainty. However, for actual RE tests, an appropriate tolerance for the pulse response should be assigned. For RE measurements, the proximity of the receiver noise floor may influence measurement results near the emission limits and its contribution is set to be 0.5 dB [26].
- The test sites: The uncertainty due to site imperfection is generally unknown. The deviation of NSA from the theoretical values [2], [3] gives an indication of the potential effect of site imperfection. However, it cannot be applied as a correction to the measured data.

  The RE measurements have been performed at ten 10 m OATSs and at two 30 m OATSs. The size of the ground plane of the 10 m OATSs is (18–23 m) $(L)\times$(8–18 m) (W). For one of the two

TABLE II
UNCERTAINTY IN RE MEASUREMENTS AT 10 m DISTANCE FOR A SPECIFIC OATS AMONG THE 12 SITES

Contributions to measurement uncertainty	Probability distribution function	Tolerance for a biconical antenna (dB)	Tolerance for a log-periodic antenna (dB)
1. Transmitting part			
a. absolute level accuracy of a signal source	rectangular	0.5	0.5
b. repeatability of the SDR	rectangular	0.5	0.5
c. mismatch between a signal source and the control unit of the SDR: ($\|\Gamma_{SG}\| = \|\Gamma_{SDR}\| = 0.3333$)	U-shaped	+0.9/-1.0	+0.9/-1.0
2. Receiving part			
a. antenna factor (AF) calibration	normal (k=2)	1.5	1.5
b. cable loss measurement	normal (k=2)	0.3	0.3
c. AF variation with height	rectangular	1.2/1.0 (H/V)	0.3/0.2 (H/V)
d. AF interpolation versus frequency	rectangular	0.3	0.3
e. antenna directivity	rectangular	+0.0/-0.0 (H) +0.2/-0.0 (V)	+0.2/-0.0 (H) +0.5/-0.0 (V)
f. antenna phase center variation	rectangular	0.0	0.2
g. cross-polarization	rectangular	0.0	0.9
h. balance	rectangular	0.3/0.9 (H/V)	0.0/0.0 (H/V)
i. mismatch between a receiving antenna and a receiver: ($\|\Gamma_{Rcvr}\| = 0.3333$)	U-shaped	+1.7/-2.1 ($\|\Gamma_{Ant}\| = 0.6429$)	+0.9/-1.0 ($\|\Gamma_{Ant}\| = 0.3333$)
j. receiver specification			
- voltage accuracy	rectangular	1.0	1.0
- absolute pulse response	rectangular	0.0	0.0
- pulse repetition response	rectangular	0.0	0.0
- noise floor proximity	normal (k=2)	0.5	0.5
3. Open area test site			
a. site imperfection	triangular	+3.2/-3.0	+3.2/-3.0
b. measurement distance variation	rectangular	0.02	0.02
c. overall system repeatability	std.dev.	0.5	0.5
Combined standard uncertainty (dB)	normal	+2.34/-2.48 (H) +2.36/-2.50 (V)	+2.07/-2.07 (H) +2.09/-2.07 (V)
Expanded uncertainty (dB)	normal (k=2)	+4.68/-4.96 (H) +4.73/-5.00 (V)	+4.15/-4.14 (H) +4.17/-4.13 (V)

30 m OATSs, the ground plane of the 10 m distance measurement is 22 m $(L)\times 18$ m (W) and for the 30 m distance measurement a narrower wire mesh of 38 m $(L)\times 8$ m (W) is extended along the measurement axis. The other 30 m OATS has the ground plane of 60 m $(L)\times 52$ m (W). The ground plane of all sites consists of metal plates and galvanized wire mesh with the spacing of less than 5.0 mm. The flatness of all sites are satisfied with the Rayleigh criterion in [29], [30]. The deviation of NSA from the theoretical values for the twelve sites are within $+1.7 - +3.7/-3.3 - -2.2$ dB and associated distribution function is assumed to be triangular. On the other hand, the NSA deviation of the 3 SACs is within 3.0 dB of the theoretical values even though the size and electromagnetic absorbers are different from site to site.

The uncertainty for distance variation is error in absolute measurement of distance between the SDR and the receive antenna. During the 10 m OATS measurement the distance was measured with the tolerance of 2.0 cm. Since the measurement distance is 10 m, the receive antenna is placed in the far-field region where the inverse-distance law is valid. Therefore, the subsequent tolerance of the electric field strength is 0.02 dB. For 3 m distance, the tolerance of 1.0 cm in measuring the distance leads to the uncertainty 0.03 dB. The overall repeatability of measurement system is typically 0.5 dB.

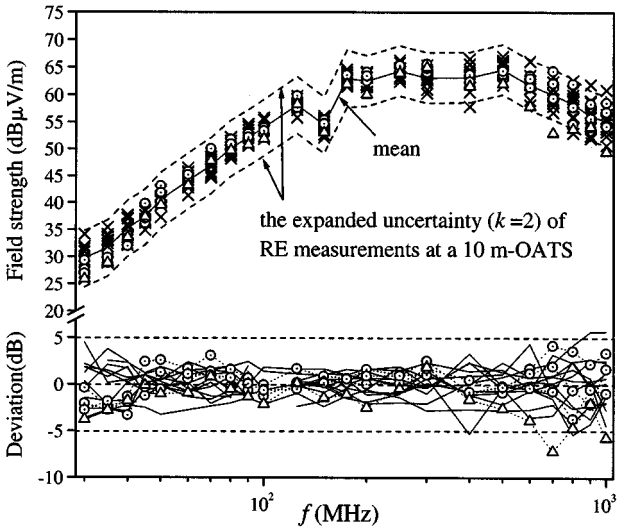


Fig. 7. The converted SAC- and measured OATS-field strengths, and their deviations from the mean of field strengths, measured at the OATSs for horizontal polarization. Field strengths: × for the OATS; ⊙ for the SAC h_2 =1–4 m), and Δ for the SAC (h_2 =1–3.54 m); deviation: — for the OATS, —⊙— for the SAC (h_2 =1–4 m), and —Δ— for the SAC (h_2 =1–3.5 m).

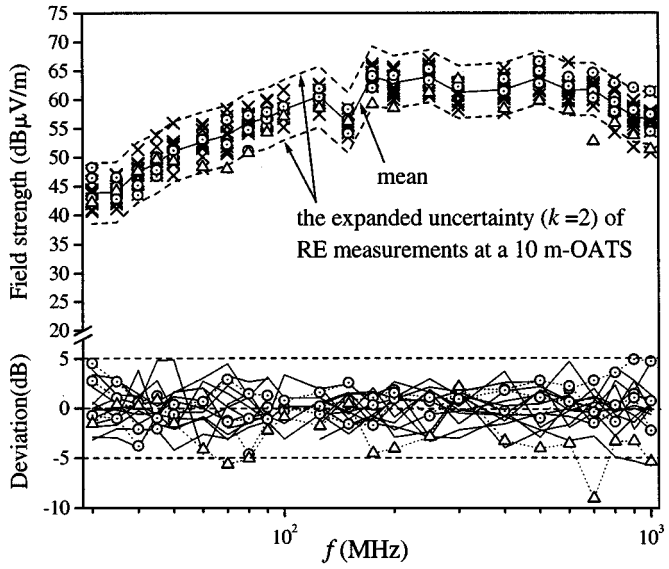


Fig. 8. The converted SAC- and measured OATS-field strengths, and their deviations from the mean of field strengths measured at the OATSs for vertical polarization. Notations are same as in Fig. 7.

V. Results and Discussion

During RE measurements not only in the 3 m SAC but at 10-m OATS, the gap voltage at each frequency has been maintained as close as possible to the reference gap voltage. Under this condition of constant radiated power, once the equivalent 10 m SAC data are obtained by applying the extrapolation factor to 3-m SAC measurement data, they can be directly compared with the measured data of the 10 m OATS. Despite the conversion of the 3-m SAC data, the standard deviation of the 3-m SAC data remains as it is.

Figs. 7 and 8 illustrate the converted 3-m SAC data and measured 10-m OATS data for horizontal and vertical polarizations, respectively. The expanded uncertainty $(k = 2)$ in RE measurements at the 10-m OATS is also drawn. The mean has been taken from averaging the data

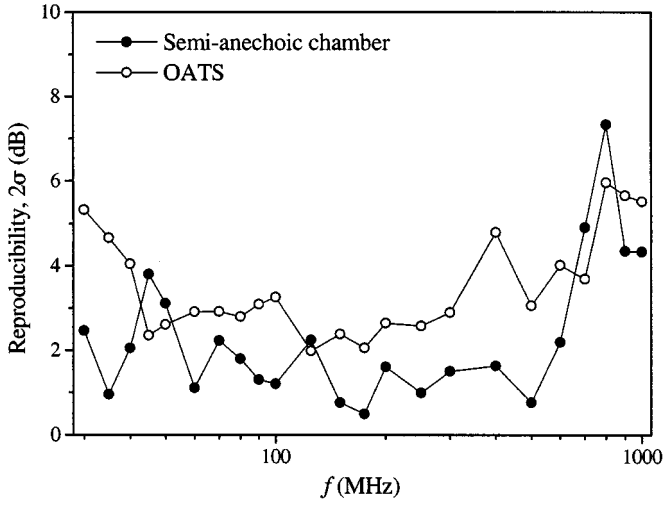


Fig. 9. Reproducibility (2σ) of RE measurements in the 3 m SACs and at 10-m OATSs for horizontal polarization.

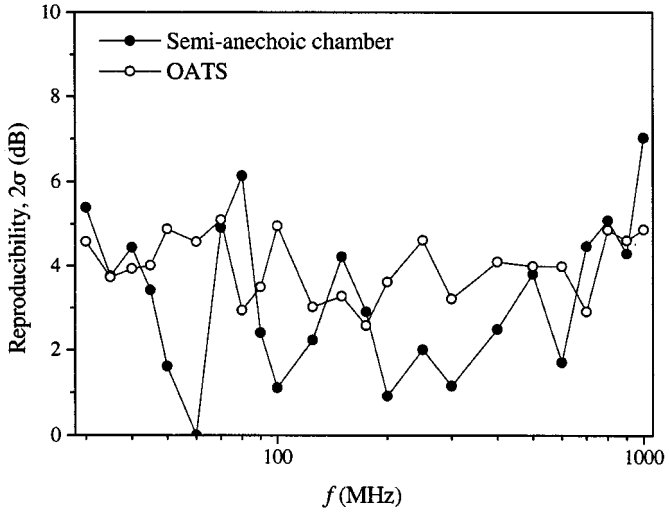


Fig. 10. Reproducibility (2σ) of RE measurements in the 3-m SACs and at 10-m OATSs for vertical polarization.

obtained at 12 OATSs. As can be observed in the two figures, it is evident that the converted SAC data are within the expanded uncertainty of the 10-m OATS measurements. Therefore, it is obvious that the methodology using the extrapolation factor for the purpose of intercomparison between data measured at different distances is experimentally verified. Furthermore, the results imply that extrapolation by the inverse distance law causes significant uncertainty.

Uncertainty in RE measurements has been evaluated in accordance with ISO Guide [27]. As previously presented, Table II shows uncertainty in RE measurements at 10 m distance for a specific OATS among the 12 sites. Uncertainty of each site depends on its instrumentation and the resultant expanded uncertainty varies site to site, but not noticeably. Among the twelve OATSs for 10 m distance, the maximum expanded uncertainty for horizontal polarization is +5.1 –5.3 dB and + 4.6 and –4.5 dB for the two frequency ranges corresponding to the biconical and log-periodic antennas. For vertical polarization, the expanded uncertainty is +5.2 –5.3 dB and + 4.6 and –4.4 dB for the two frequency ranges. This is illustrated in Figs. 7 and 8 as the upper and lower bounds from the mean.

TABLE III
REPRODUCIBILITY (2σ) OF RE MEASUREMENTS IN DECIBEL

The source	Horizontal polarization		Vertical polarization		
of data	3 m-SAC	10 m-OATS	3 m-SAC	10 m-OATS	Note
Hewlett	-	-	-	6.4	30–1000 MHz
Packard [6]				(14 OATSs)	h_1 is unknown
IBM [5]	-	-	5.0	8.6	30–500 MHz
			(17 SACs)	(21 OATSs)	$h_1 = 1$ m
Ours	7.3	6.0	7.0	5.1	30–1000 MHz
	(3 SACs)	(12 OATSs)	(3 SACs)	(12 OATSs)	$h_1 = 0.875$ m

Similarly, the maximum expanded uncertainty of 3-m SAC RE measurements for horizontally polarized waves has also been evaluated to be $+4.6/-5.0$ dB and $+4.3/-4.4$ dB for the two frequency ranges. For vertically polarized waves, the expanded uncertainty is $+4.7/-5.1$ dB and $+4.3/-4.4$ dB for the two frequency ranges. Despite closer distance in the SAC than at the OATS, the uncertainty in RE measurements both in the SAC and at the OATS shows little difference. Uncertainty analysis demonstrates that the uncertainty mainly stems from the AF calibration, mismatch, the performance of a receiver, and site imperfection.

Note in Figs. 7 and 8 that the field strengths of the SAC whose scan range of the receiving height is 1–3.5 m (denoted by the symbol of circle) are relatively lower than those measured in the other 3 SACs complied with the NSA requirement [2], [3]. It is also found that the deviation from the mean for vertical polarization is a little larger than for horizontal polarization at almost all frequencies as in [5].

Comparing the behavior versus frequency of the measured field strengths as shown in Figs. 7 and 8, the horizontal electric field is relatively lower than the vertical one in the low range of frequency. The exact theoretical behavior of horizontal and vertical electric dipoles above an ideal OATS under the condition of constant radiated power has been calculated before [13] and our results show good agreement. Here, it should be noted that the gap voltage, i.e., radiated rf power, of the SDR for both polarizations are almost same as presented in Fig. 5. However, since the sensitivity of the gap voltage of the SDR to the RF input power depends on the frequency, the resultant radiated RF power varies with the frequency. This is the reason why the curves of the mean of measured field strengths for horizontal and vertical polarizations do not vary smoothly with the frequency.

In the evaluation of reproducibility in RE measurements, we will take into account the data for the 3 SACs in which the receive antenna can be scanned in 1–4 m height range. Therefore, the number of statistically meaningful data is 3 for the SAC at all frequencies. For the OATS the number of valid data is normally 12. However, it reaches 5 at 100 MHz because of the ambient noise at OATSs, as summarized in Table I.

Figs. 9 and 10 show the evaluated reproducibility for horizontal and vertical polarizations, respectively. To compare the reproducibility behavior for the SAC and the OATS, the standard deviation is evaluated in two separated group of data; one for the SACs and the other for the OATSs. The evaluated reproducibility (2σ) is 7.3/6.0 dB and 7.0/5.1 dB for horizontal and vertical polarizations, respectively (for the SAC/for the OATS). From this, it is likely that in the viewpoint of reproducibility, the SAC is a little worse than the OATS. This results from the relatively small number of the SACs compared with the number of the OATSs and shorter distance of the SAC than the OATS. Again, note that extrapolation applied to the 3 m SAC data does not affect measurement reproducibility.

Table III summarizes reproducibility in RE measurements with other results published earlier [5], [6]. The reproducibility of RE measurements at the OATS shown in Table III is carefully taken as twice the maximum standard deviation (2σ) in the measurement frequency range discarding the data disrupted by ambient noise. Relatively inferior reproducibility of ours in the SAC originates from the small number of SACs compared with the other published results. Regarding overall range of frequency, horizontally polarized waves are more reproducible than vertically polarized ones as observed in Figs. 7 and 8 except at several frequencies.

VI. CONCLUSION

RE measurements have been made in 3 m SACs and at 10 m OATSs using an SDR as a reference emitter. An extrapolation factor analytically obtained has been employed to compare the measurement data in the SAC with those at the OATS under the different distances of the two test facilities. A commercially available SDR whose electromagnetic fields can be analytically calculated has been used as a reference emitter. The equivalent 10 m SAC data obtained by applying the extrapolation factor to the measured data at the 3 m SAC are within the expanded uncertainty $(k = 2)$ of RE measurements at the 10 m OATS. Therefore, in case of RE measurements at test facilities with different distances, the comparison of absolute field strengths using extrapolation can be used, provided that the electromagnetic property of the reference emitter is explicitly known.

Accounted possible contributions to the overall uncertainty, the expanded uncertainty $(k = 2)$ is evaluated to be 5.1 and 5.3 dB for the 3 m SAC and 10 m OATS measurements, respectively, in the frequency range of 30–1000 MHz. The reproducibility (2σ) of the RE measurements in the SAC and at the OATS has been evaluated to be 7.3 and 6.0 dB, respectively. Regarding overall range of frequency, horizontally polarized waves are more reproducible than vertically polarized ones in RE measurements.

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Off-Axis Gain and Field in the Finite Range of an Open-Ended Rectangular Waveguide

Krishnasamy T. Selvan, Dipak R. Poddar, and Kundapur R. Kini

***Abstract*—Expressions are presented for the approximate evaluation of the off-axis gain and field in the finite-range of an open-ended rectangular waveguide (OEG). These formulas in turn present a method for predicting the off-axis error when an OEG is used as the source antenna. For an OEG operating at 250 MHz, the off-axis error estimated by using this method favorably compares with that experimentally estimated for 1 m off-axis position.**

***Index Terms*—Near-field, off-axis field, off-axis gain, open rectangular waveguide.**

I. Introduction

Electromagnetic interference (EMI) and probe calibration measurements, often carried out under radiating near-field conditions in an anechoic chamber, require the generation of standard electromagnetic fields. Open-ended rectangular waveguides and pyramidal horns are among those used as transmitting antennas to generate these standard fields. The on-axis field strength is calculated as a function of range from the near-field gain of the transmitting antenna, and the on-axis wave impedance, which is normally assumed to have its free-space value [1], [2]. Though in the near-field the wave impedance would vary from its free-space value, the near-field gain is assumed to take into account the effect of this variation.

In [3] Kanda and Wyss report measurement errors in the anechoic chamber due to off-axis misalignment of the receiving antenna with respect to the source antennas (OEGs and pyramidal horns). Determining the error by comparing measured data taken on-axis and at various distances off the axis, the authors provide criteria for alignment of the receiving probes in their report.

In this paper, a theoretical approach is presented for the approximate estimation of the off-axis error when an open-ended rectangular waveguide is used as the source antenna. This approach comprises derivation of approximate formulas for the off-axis gain of an OEG in the E- and H-planes, and the subsequent employment of a field strength formula [2].

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